

4 (a) Taking downslope as positive, $mg \sin \theta - T = ma$ [1]

At equilibrium position: $mg \sin \theta - T = 0$
 $mg \sin \theta - ke = 0$
 $mg \sin \theta = ke$ [1]

At further extension x : $mg \sin \theta - k(e+x) = ma$
 $mg \sin \theta - ke - kx = ma$
 $ke - ke - kx = ma$ [1]
 $a = -\frac{k}{m}x$

(b) k and m are constants, therefore a is proportional to x . [1]
Negative sign shows that a is always opposite to x [1]

(c) $v = \pm \omega \sqrt{x_0^2 - x^2}$

when $v = 0.25 v_0$, $0.25v_0 = \omega \sqrt{x_0^2 - x^2}$
 $0.25(x_0\omega) = \omega \sqrt{x_0^2 - x^2}$ [1]
 $0.25(3.0) = \sqrt{3.0^2 - x^2}$
 $x^2 = 8.25$
 $x = 2.87 = 2.9 \text{ cm}$ [1]

(d) (i) The driving frequency of the oscillator is equal to the natural frequency of the mass-spring system, resulting in resonance. [1]

(maximum energy is transferred to the system) and the mass-spring system oscillates with maximum amplitude, [1]

(ii) $a_0 = x_0\omega^2 = (0.060)(2\pi \times 1.2)^2 = 3.41 = 3.4 \text{ m s}^{-2}$ [1]

